

$\therefore$ Smith lies and Smith is guilty

$\therefore$ If Jones is innocent, then both Smith and Williams are guilty. But this is a contradiction to the fact that exactly one person must be guilty.

$\therefore$ Jones is not innocent and Jones is guilty of murder

37) Steve would like to determine the relative salaries of three coworkers using two facts. First, he knows that if Fred is not the highest paid of the three, then Janice is. Second, he knows that if Janice is not the lowest paid, then Maggie is paid the most. Is it possible to determine the relative salaries of Fred, Maggie, and Janice from what Steve knows? If so, who is paid the most, and who the least? Explain your reasoning.

Soln: p : Fred is the highest paid q : Janice is the highest paid ($q \Leftrightarrow \neg s$ is True)
 r : Maggie is paid the most s : Janice is the lowest paid

$\neg p \Rightarrow q$ is True $\neg s \Rightarrow r$ is True ~~$\neg p \Rightarrow r$~~ $r \Rightarrow \neg p$ is True

Let $\neg s$ be True. By defn. of conditional, r is True. By defn. of conditional, $\neg p$ is True. By defn. of conditional, q is True. By defn. of conditional, $\neg p$ is True. This is a contradiction since p is True and by defn. of negation, $\neg p$ should be False.

$q \Rightarrow \neg r$ is True. $\therefore \neg s$ is False and by defn. of negation, s is True. $\therefore$ Janice is the lowest paid of the three

Let $\neg p$ be True. By defn. of conditional, q is True. By defn. of bi-conditional, $\neg s$ is True. This is a contradiction.

$\therefore \neg p$ is False, and by defn. of negation, p is True. $\therefore$ Fred is the highest paid

$\therefore$ Fred > Maggie > Janice in terms of salary.